UCS531-Cloud Computing

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Amazon Web Services

What is AWS?

The Amazon Web Services (AWS) service is provided by the Amazon that uses distributed IT infrastructure to provide different IT resources available on demand.

It provides different services such as infrastructure as a service (IaaS), platform as a service (PaaS) and packaged software as a service (SaaS).

Amazon launched AWS, a cloud computing platform to allow the different organizations to take advantage of reliable IT infrastructure.

AWS Geographical Regions

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AWS is located in 9 geographical 'Regions

Each Region is wholly contained within a single country and all of its data and services stay within the designated Region.

Each Region has multiple 'Availability Zones', which are distinct data centers providing AWS services.

Availability Zones are isolated from each other to prevent outages from spreading between Zones.

However, Several services operate across Availability Zones (e.g. S3, DynamoDB).

What AWS Offers?

Low Ongoing Cost: pay-as-you-go pricing with no up-front expenses or long-term commitments.

Instant Elasticity & Flexible Capacity: (scaling up and down) Eliminate guessing on your infrastructure capacity needs.

Speed & Agility: Develop and deploy applications faster instead of waiting weeks or months for hardware to arrive and get installed.

Apps not Ops: Focus on projects. Lets you shift resources away from data center investments and operations and move them to innovative new projects.

Global Reach: Take your apps global in minutes. Open and Flexible: You choose the development platform or programming model that makes the most sense for your business.

Secure: Allows your application to take advantage of the multiple layers of operational and physical security in the AWS data centers to ensure the integrity and safety of your data.

Uses of AWS

A small manufacturing organization uses their expertise to expand their business by leaving their IT management to the AWS.

A large enterprise spread across the globe can utilize the AWS to deliver the training to the distributed workforce.

An architecture consulting company can use AWS to get the high-compute rendering of construction prototype.

A media company can use the AWS to provide different types of content such as ebox or audio files to the worldwide files.

Pay-As-You-Go

Based on the concept of Pay-As-You-Go, AWS provides the services to the customers.

AWS provides services to customers when required without any prior commitment or upfront investment.

Pay-As-You-Go

enables the customers to procure services from AWS.

- Computing
- Programming models
- Database storage
- Networking

Availability Zones

An Availability Zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity in an AWS Region.

AZs give customers the ability to operate production applications and databases that are more highly available, fault tolerant, and scalable than would be possible from a single data center.

All AZs in an AWS Region are interconnected with high-bandwidth, low-latency networking, over fully redundant, dedicated metro fiber providing high-throughput, low-latency networking between AZs.

Advantages of AWS

1) Flexibility

- We can get more time for core business tasks due to the instant availability of new features and services in AWS.
- It provides effortless hosting of legacy applications.
- AWS does not require learning new technologies and migration of applications to the AWS provides the advanced computing and efficient storage.
- AWS also offers a choice that whether we want to run the applications and services together or not.
- We can also choose to run a part of the IT infrastructure in AWS and the remaining part in data centres.

2) Cost-effectiveness

• Traditional IT infrastructure that requires a huge investment.

• AWS requires no upfront investment, long-term commitment, and minimum expense.

3) Scalability/Elasticity

•Through AWS, autoscaling and elastic load balancing techniques are automatically scaled up or down, when demand increases or decreases respectively.

• AWS techniques are ideal for handling unpredictable or very high loads.

• Due to this reason, organizations enjoy the benefits of reduced cost and increased user satisfaction.

4) Security

• AWS provides end-to-end security and privacy to customers.

•AWS has a virtual infrastructure that offers optimum availability while managing full privacy and isolation of their operations.

• Customers can expect high-level of physical security because of Amazon's several years of experience in designing, developing and maintaining large-scale IT operation centers.

• AWS ensures the three aspects of security, i.e., Confidentiality, integrity, and availability of user's data.

AWS Global Infrastructure

• Global infrastructure is a region around the world in which AWS is based.

• AWS is available in 19 regions, and 57 availability zones in December 2018 and 5 more regions 15 more availability zones for 2019.

AWS Cloud spans 120 Availability Zones within 38 Geographic Regions, with announced plans for 10 more Availability Zones and 3 more AWS Regions in the Kingdom of Saudi Arabia, Chile, and the AWS European Sovereign Cloud.

AWS Geographical Regions

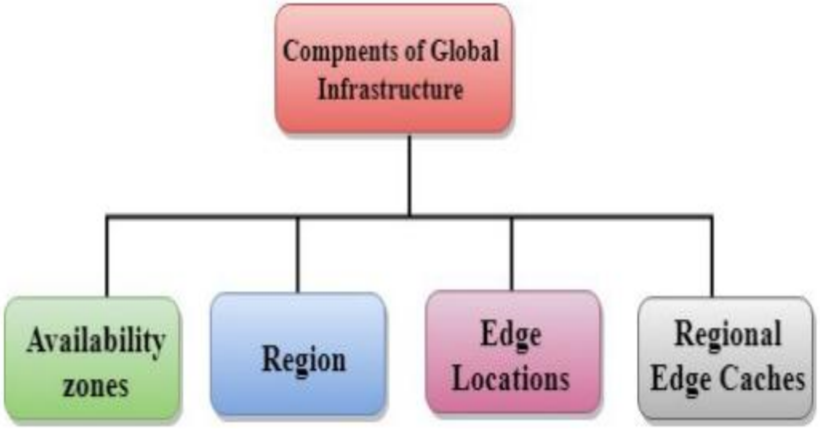
AWS opens new Regions rapidly.

AWS maintains multiple geographic Regions, including Regions in North America, South America, Europe, China, Asia Pacific, South Africa, and the Middle East.

- The following are the components that make up the AWS infrastructure:
  - Availability Zones
- Region
- Edge locations
- Regional Edge Caches

Availability zone as a Data Center

- An availability zone is a facility that can be somewhere in a country or in a city.
- Inside this facility, *i.e.*, Data Centre, we can have multiple servers, switches, load balancing, firewalls.
- The things which interact with the cloud sits inside the data centers.
- An availability zone can be a several data centers, but if they are close together, they are counted as 1 availability zone.



Region

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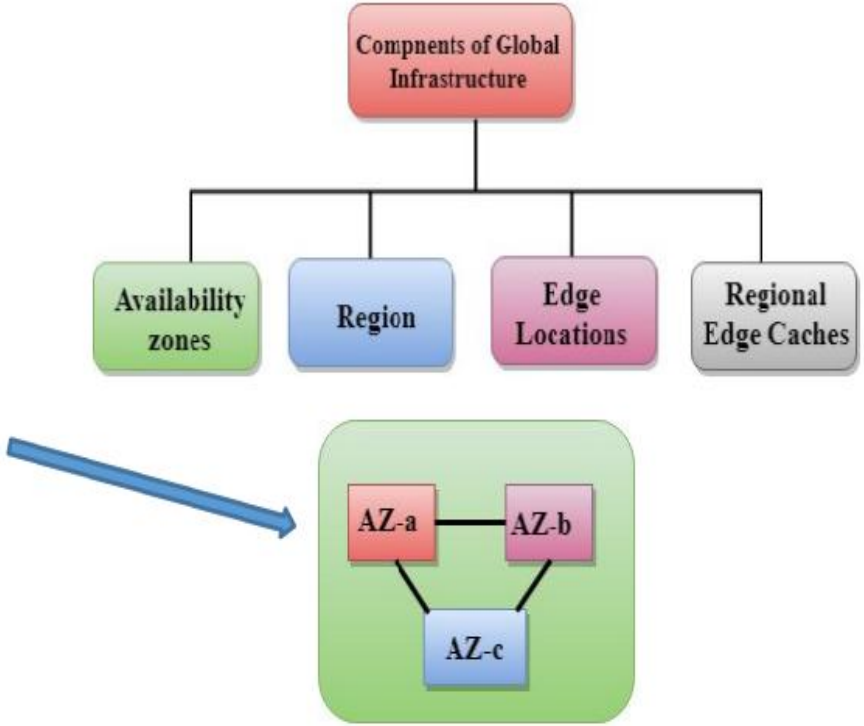
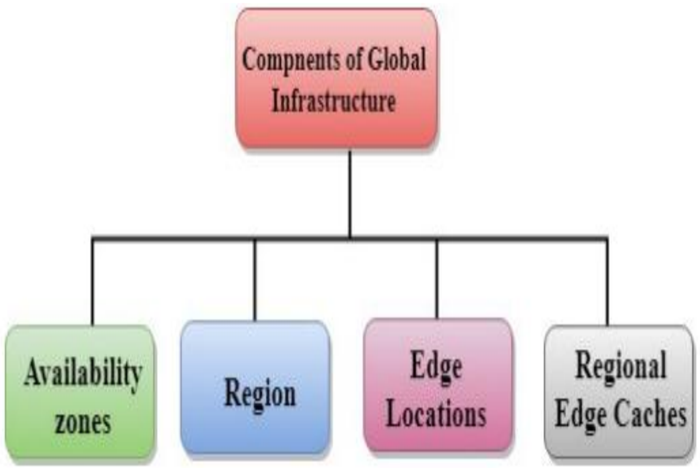
A region is a geographical area.

- Each region consists of 2 more availability zones.
- A region is a collection of data centers which are completely isolated from other regions.
- A region consists of more than two availability zones connected to each other through links.
- Availability zones are connected through redundant and isolated metro fibers.



Edge Locations

- **Edge locations** are the endpoints for AWS used for caching content.
- Edge locations consist of CloudFront, Amazon's Content Delivery Network (CDN).
- **Edge location** is not a region but a small location that AWS have.
- Located in most of the major cities.
- For example, some user accesses your website from Singapore; then this request would be redirected to the edge location closest to Singapore where cached data can be read.



Regional Edge Cache

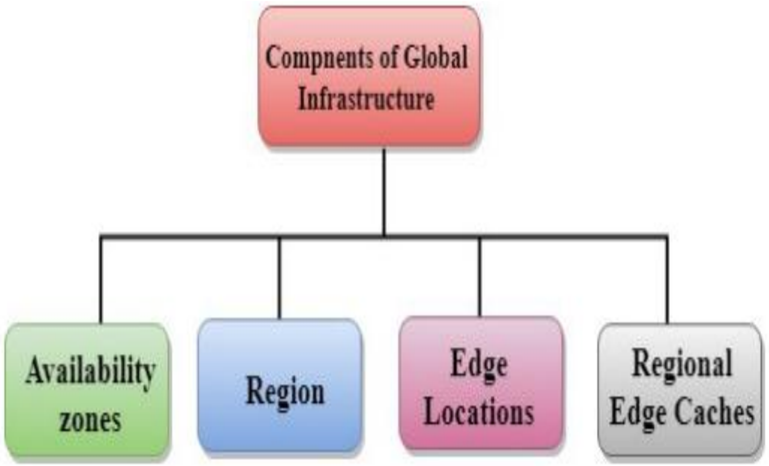
• AWS announced a new type of edge location in November 2016, known as a Regional Edge Cache.

• Regional Edge cache lies between Cloud-Front Origin servers and the edge locations.

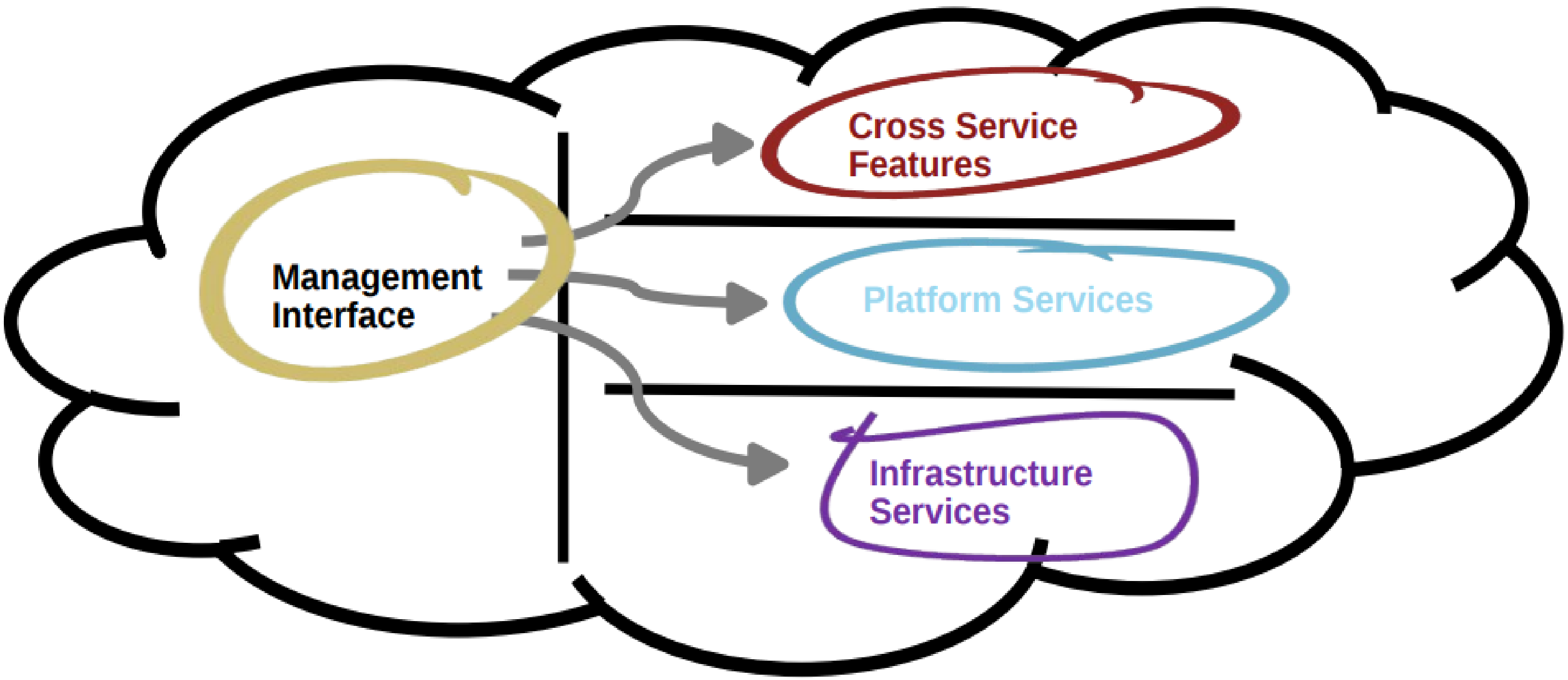
• A regional edge cache has a large cache than an individual edge location.

• Data is removed from the cache at the edge location while the data is retained at the Regional Edge Caches.

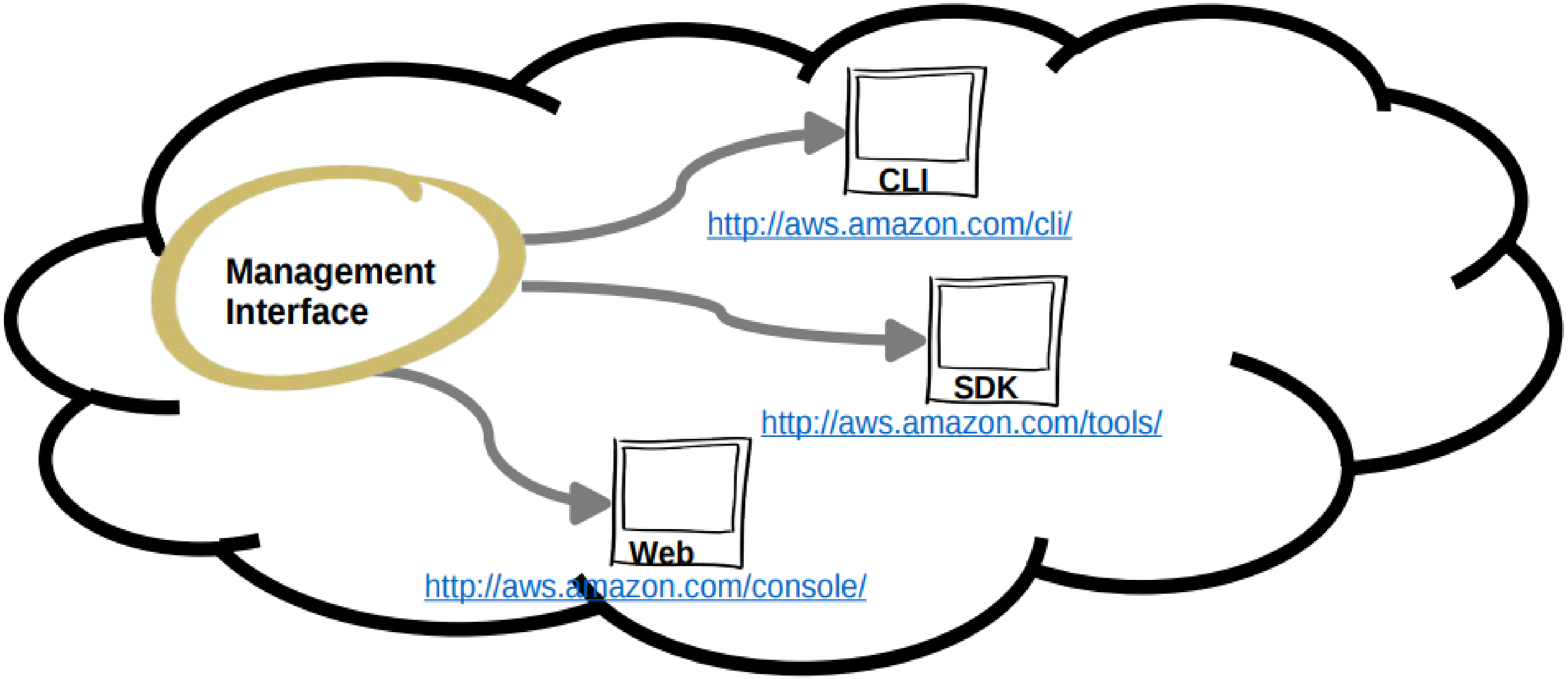
• **Edge location retrieves the cached data from the Regional edge cache instead of the Origin servers that have high latency**.



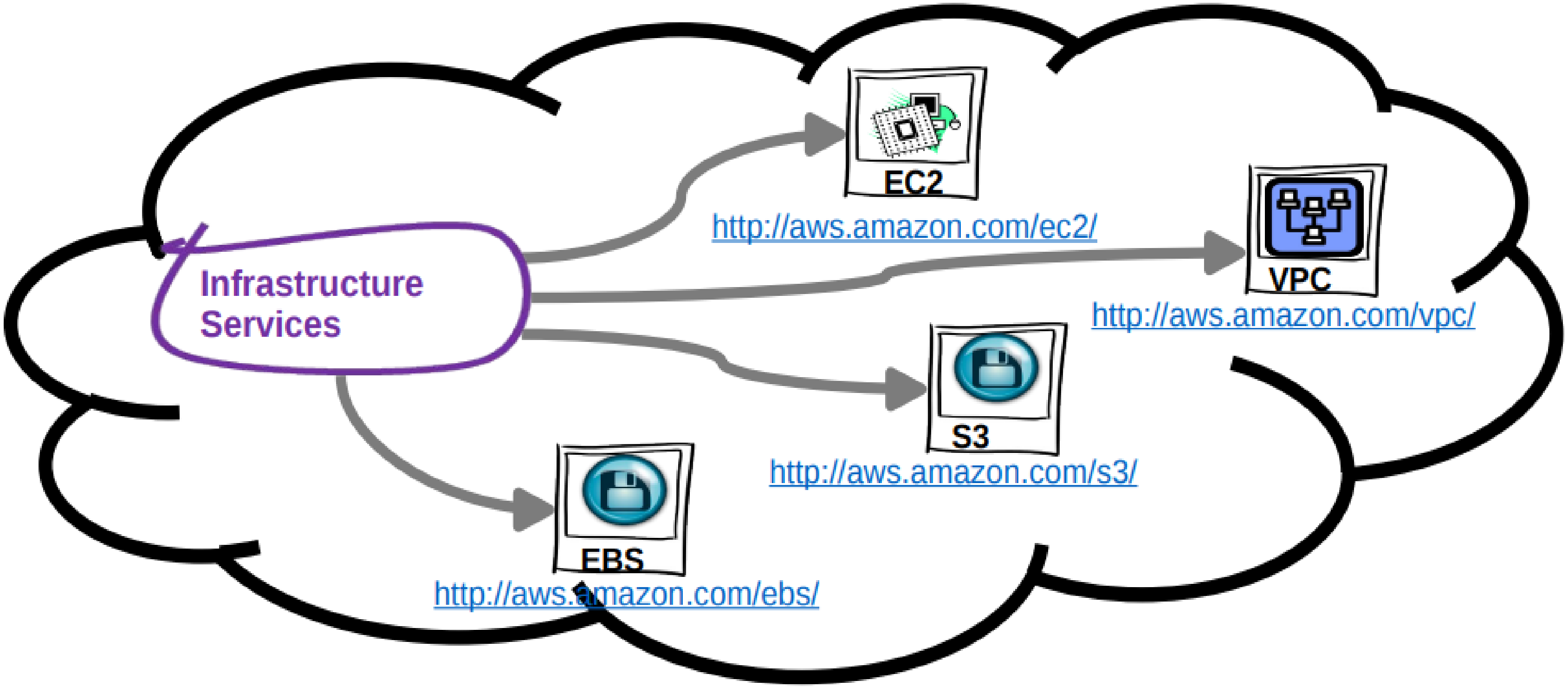
The Amazon Web Services Universe



Management Interface



Infrastructure Services



Amazon Elastic Compute Cloud (EC2)

A web service that provides resizable compute capacity in the cloud.

EC2 allows creating Virtual Machines (VM) on-demand. Pre-configured templated Amazon Machine Image (AMI) can be used get running immediately. Creating and sharing your own AMI is also possible via the AWS Marketplace.

- **Auto Scaling** allows automatically scale of the capacity up seamlessly during demand spikes to maintain performance, and scales down during demand lulls to minimize costs.
- **Elastic Load Balancing** automatically distributes incoming application traffic across multiple Amazon EC2 instances.
- Provide **tools to build failure resilient applications** by launching application instances in separate Availability Zones.
- Pay only for resources actually consume, instance-hours.
- **VM Import/Export** enables you to easily import virtual machine images from your existing environment to Amazon EC2 instances and export them back at any time.

EC2 Instances

Micro instances (t1.micro): – Micro Instance 613 MiB of memory, up to 2 ECUs (for short periodic bursts), EBS storage only, 32-bit or 64-bit platform.

▪ Standard Instances provide customers with a balanced set of resources and a low cost platform.

– M1 Small Instance (Default) 1.7 GiB of memory, 1 EC2 Compute Unit (1 virtual core with 1 EC2 Compute Unit), 160 GB of local instance storage, 32-bit or 64-bit platform

– M1 Medium Instance 3.75 GiB of memory, 2 EC2 Compute Units (1 virtual core with 2 EC2 Compute Units each), 410 GB of local instance storage, 32-bit or 64-bit platform

– M1 Large Instance 7.5 GiB of memory, 4 EC2 Compute Units (2 virtual cores with 2 EC2 Compute Units each), 850 GB of local instance storage, 64-bit platform

– M1 Extra Large Instance 15 GiB of memory, 8 EC2 Compute Units (4 virtual cores with 2 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform

– M3 Extra Large Instance 15 GiB of memory, 13 EC2 Compute Units (4 virtual cores with 3.25 EC2 Compute Units each), EBS storage only, 64-bit platform

– M3 Double Extra Large Instance 30 GiB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), EBS storage only, 64-bit platform

One EC2 Compute Unit (ECU) provides the equivalent CPU capacity of a 1.0-1.2 GHz 2007 Opteron or 2007 Xeon processor

EC2 High Performance Instances

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▪ **High-Memory Instances**: – High-Memory Extra Large Instance 17.1 GiB memory, 6.5 ECU (2 virtual cores with 3.25 EC2 Compute Units each), 420 GB of local instance storage, 64-bit platform

– **High-Memory Double Extra Large Instance** 34.2 GiB of memory, 13 EC2 Compute Units (4 virtual cores with 3.25 EC2 Compute Units each), 850 GB of local instance storage, 64-bit platform

– **High-Memory Quadruple Extra Large Instance** 68.4 GiB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform ▪ High-CPU Instances

– **High-CPU Medium Instance** 1.7 GiB of memory, 5 EC2 Compute Units (2 virtual cores with 2.5 EC2 Compute Units each), 350 GB of local instance storage, 32-bit or 64-bit platform

– **High-CPU Extra Large Instance** 7 GiB of memory, 20 EC2 Compute Units (8 virtual cores with 2.5 EC2 Compute Units each), 1690 GB of local instance storage, 64-bit platform ▪ High Storage Instances

– **High Storage Eight Extra Large** 117 GiB memory, 35 EC2 Compute Units, 24 * 2 TB of hard disk drive local instance storage, 64-bit platform, 10 Gigabit Ethernet ▪ High I/O Instances

– **High I/O Quadruple Extra Large** 60.5 GiB memory, 35 EC2 Compute Units, 2 * 1024 GB of SSD-based local instance storage, 64-bit platform, 10 Gigabit Ethernet

EC2 Cluster Instances

▪ **Cluster Compute Instances** provide proportionally high CPU resources with increased network performance and are well suited for High Performance Compute (HPC) applications and other demanding network-bound applications.

– **Cluster Compute Eight Extra** Large 60.5 GiB memory, 88 EC2 Compute Units, 3370 GB of local instance storage, 64-bit platform, 10 Gigabit Ethernet

▪ **High Memory Cluster Instances** provide proportionally high CPU and memory resources with increased network performance, and are well suited for memory-intensive applications including in-memory analytics, graph analysis, and scientific computing.

– **High Memory Cluster Eight Extra** Large 244 GiB memory, 88 EC2 Compute Units, 240 GB of local instance storage, 64- bit platform, 10 Gigabit Ethernet

▪ **Cluster GPU Instances** provide general-purpose graphics processing units (GPUs) with proportionally high CPU and increased network performance for applications benefitting from highly parallelized processing, including HPC, rendering and media processing applications.

– **Cluster GPU Quadruple Extra** Large 22 GiB memory, 33.5 EC2 Compute Units, 2 x NVIDIA Tesla "Fermi" M2050 GPUs, 1690 GB of local instance storage, 64-bit platform, 10 Gigabit Ethernet.

EC2 Payment methods

- **On-Demand Instances** let you pay for compute capacity by the hour with no long-term commitments.
- **Reserved Instances** give you the option to make a low, one-time payment for each instance you want to reserve and in turn receive a significant discount on the hourly charge for that instance.
- **Spot Instances** allow customers to bid on unused Amazon EC2 capacity and run those instances for as long as their bid exceeds the current Spot Price

Storage is another AWS core service.

There are three broad categories of storage:
instance store ("ephemeral"), Amazon Elastic Block Store (EBS), and Amazon Simple Storage Service (S3).

Instance store, or ephemeral storage, is a temporary storage that it is added to the Amazon EC2 instance.

Amazon EBS is persistent, mountable storage, which can be mounted as a device to an Amazon EC2 instance. Amazon EBS can only be mounted to an Amazon EC2 instance within the same Availability Zone.
Similar to Amazon EBS, Amazon S3 is persistent storage; however, it can be accessed from anywhere.

Core AWS Services

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Amazon VPC

Amazon EC2

Amazon S3

Amazon EBS

Amazon EFS

Amazon Glacier

Storage

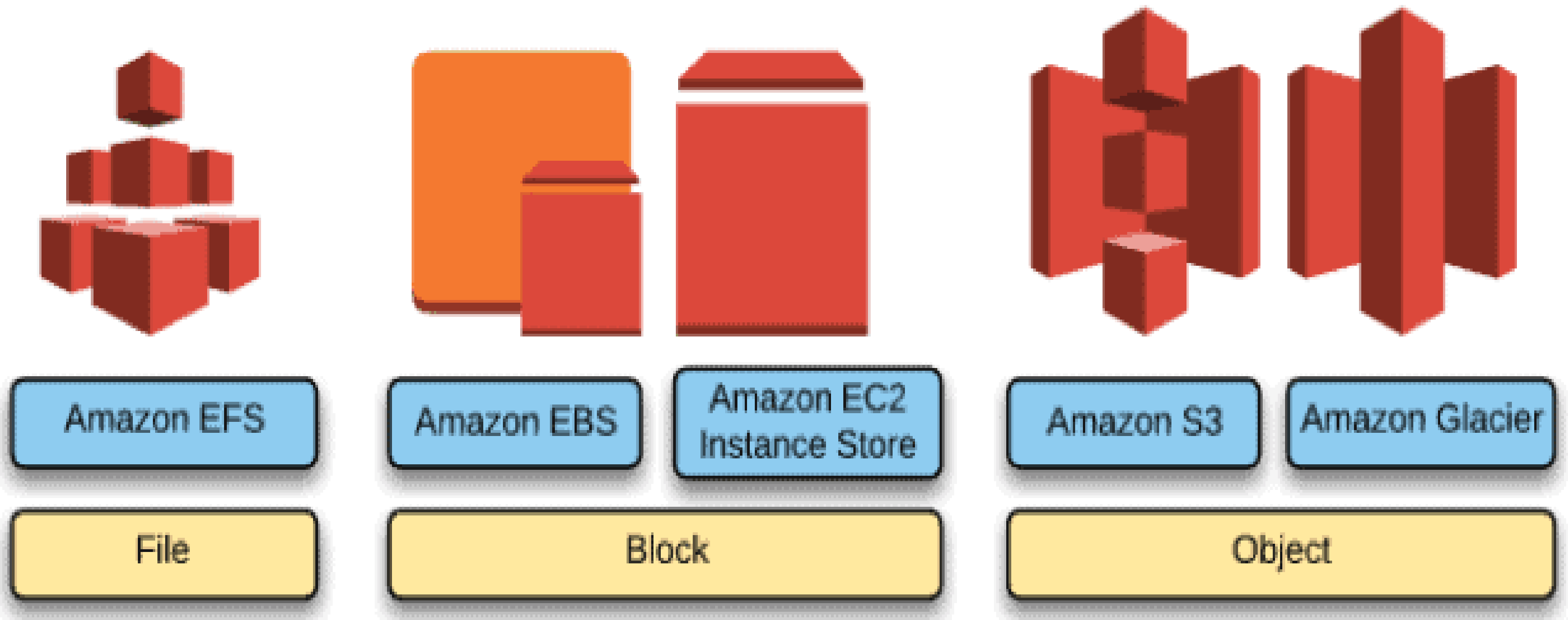
Amazon RDS

Amazon DynamoDB

Database

AWS IAM

AWS storage choices



Amazon Elastic Block Store (Amazon EBS)

It offers persistent storage for Amazon EC2 instances. Amazon EBS volumes are network-attached and persist independently from the life of an instance.

Amazon EBS volumes are highly available, highly reliable volumes that can be leveraged as an Amazon EC2 instances boot partition or attached to a running Amazon EC2 instance as a standard block device.

Amazon Elastic Block Store (EBS)

▪ **Provides block level storage volumes** (1 GB to 1 TB) for use with Amazon EC2 instances. – Multiple volumes can be mounted to the same instance.

– **EBS volumes are network-attached**, and persist independently from the life of an instance.

– **Storage volumes behave like raw**, unformatted block devices, allowing users to create a file system on top of Amazon EBS volumes, or use them in any other way you would use a block device (like a hard drive).

▪ **EBS volumes** are placed in a specific Availability Zone, and can then be attached to instances also in that same Availability Zone.

▪ Each storage volume is automatically replicated within the same Availability Zone.

▪ **EBS provides the ability to create point-in-time snapshots of volumes, which are persisted to Amazon S3.**

– These snapshots can be used as the starting point for new Amazon EBS volumes, and protect data for long-term durability.

– The same snapshot can be used to instantiate as many volumes as you wish. – These snapshots can be copied across AWS regions.

EBS Volumes

▪ **Standard volumes** offer storage for applications with moderate or bursty I/O requirements.

– Standard volumes deliver approximately 100 IOPS on average.

– well suited for use as boot volumes, where the burst capability provides fast instance start-up times.

▪ **Provisioned IOPS volumes** are designed to deliver predictable, high performance for I/O intensive workloads such as databases.

– You specify an IOPS rate when creating a volume, and EBS provisions that rate for the lifetime of the volume.

– Amazon EBS currently supports up to 4000 IOPS per Provisioned IOPS volume.

– You can stripe multiple volumes together to deliver thousands of IOPS per EC2 instance.

▪ To enable your EC2 instances to fully utilize the IOPS provisioned on an EBS volume,:

– Launch selected Amazon EC2 instance types as "EBS-Optimized" instances. – EBS-optimized instances deliver dedicated throughput between Amazon EC2 and Amazon EBS, with options between 500 Mbps and 1000 Mbps depending on the instance type used.

▪ **EBS charges based on per GB-month AND per 1 million I/O requests**

Amazon EBS Volume Features

Amazon EBS volumes deliver the following features:

- **Persistent storage:** Volume lifetime is independent of any particular Amazon EC2 instance.
- **General purpose:** Amazon EBS volumes are raw, unformatted block devices that can be used from any operating system.
- **High performance:** Amazon EBS volumes are equal to or better than local Amazon EC2 drives.
- **High reliability:** Amazon EBS volumes have built-in redundancy within an Availability Zone.
- **Designed for resiliency:** The AFR (Annual Failure Rate) of Amazon EBS is between 0.1% and 1%.
- **Variable size:** Volume sizes range from 1 GB to 16 TB.
- **Easy to use:** Amazon EBS volumes can be easily created, attached, backed up, restored, and deleted.

Block Storage

- Suitable for transactional db, random read/write, and structured db storage
- Data is divided and stored in evenly sized blocks.
- Data blocks would not contain metadata.
- It only contains index of data block, doesn't care about the data in it.

Object Storage

- Stores the files as a whole and doesn't divide them.
- An object has its data and metadata with a unique ID
- It can not be mounted as a drive
- Global unique ID is unique globally and it can be retrieved globally.

Amazon EBS Volume Types

	Solid-State Drives (SSD)		Hard Disk Drives (HDD)	
	General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Max volume size	16 TiB	16 TiB	16 TiB	16 TiB
Max IOPS/volume	16,000	64,000	500	250
Max throughput/volume	250 MiB/s	1,000 MiB/s	500 MiB/s	250 MiB/s

Amazon EBS Volume Types

	Solid-State Drives (SSD)		Hard Disk Drives (HDD)	
	General Purpose	Provisioned IOPS	Throughput-Optimized	Cold
Use Cases	• Recommended for most workloads • System boot volumes • Virtual desktops • Low-latency interactive apps • Development and test environments	• Critical business applications that require sustained IOPs performance, or more than 16,000 IOPS or 250 MiB/s of throughput per volume • Large database workloads	• Streaming workloads requiring consistent, fast throughput at a low price • Big data • Data warehouses • Log processing • Cannot be a boot volume	• Throughput-oriented storage for large volumes of data that is infrequently accessed • Scenarios where the lowest storage cost is important • Cannot be a boot volume

Amazon Simple Storage Service (S3)

- Amazon S3 provides a simple web services interface that can be used to store and retrieve any amount of data, at any time, from anywhere on the web.
- Write, read, and delete objects containing from 1 byte to 5 terabytes of data each. The number of objects you can store is unlimited.
- **Each object is stored in a bucket** and retrieved via a unique, developer-assigned key.

– A bucket can be stored in one of several Regions.

– Choose a Region to optimize for latency, minimize costs, or address regulatory requirements.

– Objects stored in a Region never leave the Region unless you transfer them out.

- Authentication mechanisms are provided to ensure that data is kept secure from unauthorized access.

– Objects can be made private or public, and rights can be granted to specific users.

- S3 charges based on per GB-month AND per I/O requests AND per data modification requests.

Amazon Simple Storage Service (Amazon S3)

Amazon Simple Storage Service (Amazon S3) is an **object storage service** that offers industry-leading scalability, data availability, security, and performance.

The **customers of all sizes**, organizations and industries can use it to store and protect any amount of data.

S3 are used for **variety of uses** like websites, mobile applications, backup and restore, archive, enterprise applications, IoT devices, and big data analytics.

Amazon S3 provides **easy-to-use** management **features**.

Features S3

Unlimited number of objects with **write, read, and delete** facility and containing 1 byte to 5 terabytes of data.

Each object is stored in a **bucket** and retrieved via a **unique**, developer-assigned **key**.

Objects stored in a specific **region never leave** the Region unless transferred.

Authentication mechanisms are provided to ensure that data is kept secure from unauthorized access.

Uses **standards**-based REST and SOAP interfaces to work with any Internet-development toolkit.

Built to be **flexible** so that protocol or functional layers can easily be added.

Provides functionality to simplify manageability of data through its lifetime.

Starting with S3

Launching an Amazon S3 service for the **first time** need the following steps:

- Sign up for AWS
- Create an IAM (Identity and Access Management) user
- Sign in as an IAM user
- Creating a bucket
- Uploading an object to a bucket
- Viewing an object
- Deleting objects and buckets

Sign Up with S3

To create an AWS account

Open https://portal.aws.amazon.com/billing/signup.

Follow the online instructions.

Create an IAM user with administrative privilege.

Create a bucket

To create a bucket

Sign in to the AWS, open S3 console https://console.aws.amazon.com/s3/

Choose **Create bucket**.

In **Bucket name**, enter a name for your bucket with following constraints:
Be unique across all of Amazon S3.
Be between 3 and 63 characters long.
Not contain uppercase characters.
Start with a lowercase letter or number.

After you create the bucket, you can't change its name.

In Region, choose the AWS Region where you want the bucket to reside.

In Bucket settings for Block Public Access, keep the values set to the defaults.

Choose Create bucket.

Upload an object to a bucket

To upload an object to a bucket

In the Bucket list, select the bucket that you want to upload your object to.

On the Overview tab for your bucket, choose Upload or Get Started.

In the Upload dialog box, choose Add files.

Choose a file to upload, and then choose Open.

Choose Upload.

To download an object from a bucket

In the **Buckets** list, choose the name of the bucket that you created.

In the **Name** list, choose the name of the object that you uploaded.

For your selected object, the object overview panel opens.

On the **Overview** tab, review information about your object.

To view the object in your browser, choose **Open**.

To download the object to your computer, choose **Download**.

Delete an object and Bucket

Delete an object.

In the **Buckets** list, choose the bucket that you want to delete an object from.

In the **Name** list, select the check box for the object that you want to delete.

Choose **Actions**, and then choose **Delete**.

In the **Delete objects** dialog box, verify that the name of the object, and choose **Delete**.

Delete your bucket.

To delete a bucket, in the **Buckets** list, select the bucket.

Choose **Delete**.

Securing Amazon S3 buckets and objects

Newly created S3 buckets and objects are private and protected by default

When use cases must share Amazon S3 data –

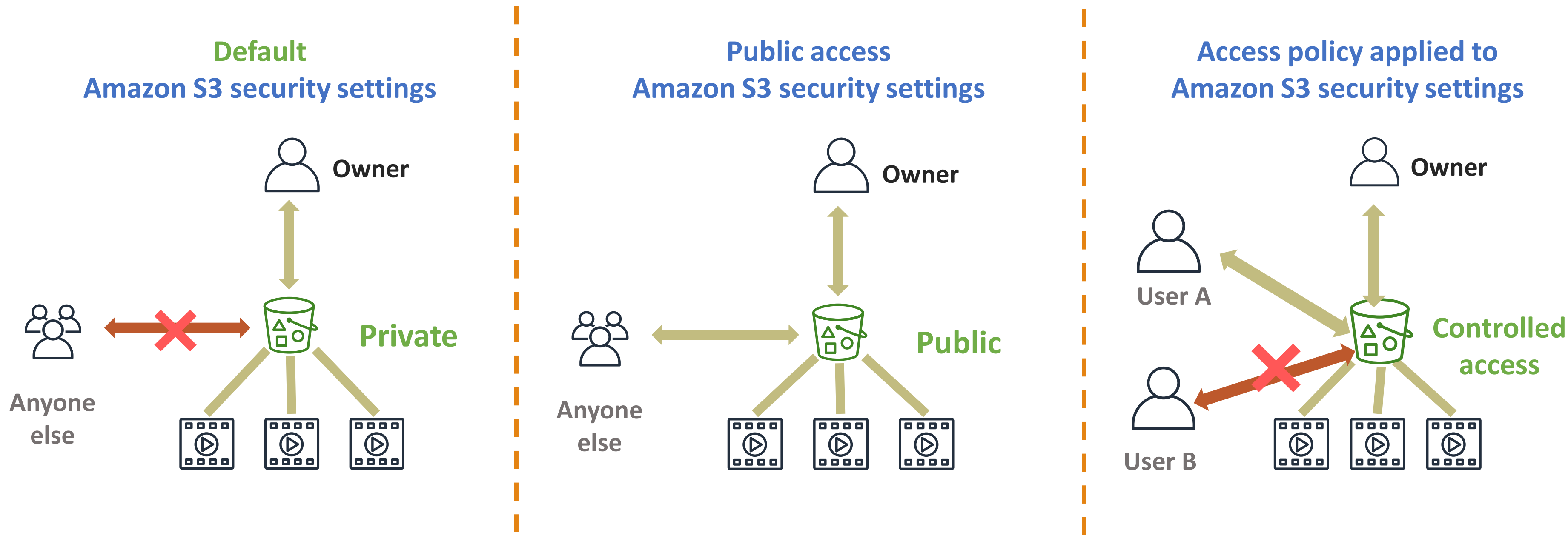
- Manage and control the data access
- Follow the principle of least privilege

Tools and options for controlling access to Amazon S3 data –

- Block Public Access feature: It is enabled on new buckets by default, simple to manage
- IAM policies: A good option when the user can authenticate using IAM
- Bucket policies: You can define access to a specific object or bucket
- Access control lists (ACLs): A legacy access control mechanism
- S3 Access Points: You can configure access with names and permissions specific to each application
- Presigned URLs: You can grant time-limited access to others with temporary URLs
- AWS Trusted Advisor bucket permission check: A free feature

Three general approaches to configuring access

Configure the appropriate security settings for your use case on the bucket and objects.



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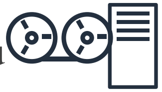
Consider encrypting objects in Amazon S3

Encryption encodes data with a secret key, which makes it unreadable

- Only users who have the secret key can decode the data
- Optionally, use AWS Key Management Service (AWS KMS) to manage secret keys

Server-side encryption

- On the bucket, enable this feature by selecting the Default encryption option
- Amazon S3 encrypts objects before it saves the objects to disk, and decrypts the objects when you download them



Client-side encryption

- Encrypt data on the client side and upload the encrypted data to Amazon S3
- In this case, you manage the encryption process

Amazon S3 benefits

Durability

- It ensures data is not lost
- S3 Standard storage provides 11 9s (or 99.999999999%) of durability

Availability

- You can access your data when needed
- S3 Standard storage class is designed for four 9s (or 99.99%) availability

Scalability

- It offers virtually unlimited capacity
- Any single object of 5 TB or less

Security

- It offers fine-grained access control

Performance

- It is supported by many design patterns

Amazon S3 provides many features that make it an important component of many solutions built on AWS.

First, it provides *durability*, which describes the average annual expected loss of objects. 11 9s of durability means that every year, there is a 0.000000001 percent chance of losing an object. For example, if you store 10,000 objects on Amazon S3, you can expect to incur a loss of a single object once every 10,000,000 years on average. Amazon S3 redundantly stores your objects on multiple devices across multiple facilities in the Amazon S3 Region you designate. Amazon S3 is designed to sustain concurrent device failures by quickly detecting and repairing any lost redundancy. Amazon S3 also regularly verifies the integrity of your data by using checksums.

Amazon S3 also provide four 9s (or 99.99 percent) of *availability*. Availability refers to your ability to access your data quickly, when you want it. It also provides a virtually unlimited capacity to store your data, so it is *scalable*. Amazon S3 has robust *security* settings. It provides many ways to control access to the data that you store, and also enables you to encrypt your data.

Finally, Amazon S3 is *highly performant*, with a first-byte latency that is measured in milliseconds for most storage classes. For more information about S3 performance design patterns, see the Amazon S3 Documentation. Common approaches include using caching for frequently accessed content; configurable retry and timeout logic for objects that receive significant request traffic in a short period of time; and horizontal scaling and request parallelization for high throughput across the network.

S3 Storage pricing

For storing objects in the S3 buckets.

The rate are charged depends on the objects' size, how long stored the objects during the month, and the storage class—

S3 Standard,

S3 Intelligent-Tiering

S3 Standard-Infrequent Access

S3 One Zone-Infrequent Access

S3 Express One Zone

S3 Glacier Instant Retrieval

S3 Glacier Flexible Retrieval (Formerly S3 Glacier)

S3 Glacier Deep Archive.

You pay a monthly monitoring and automation charge per object stored in the S3 Intelligent-Tiering storage class to monitor access patterns and move objects between access tiers. In S3 Intelligent-Tiering there are no retrieval charges, and no additional tiering charges apply when objects are moved between access tiers.

There are per-request ingest charges when using PUT, COPY, or lifecycle rules to move data into any S3 storage class. Consider the ingest or transition cost before moving objects into any storage class.

Estimate costs using the AWS Pricing Calculator.

Feature	AWS S3 (Simple Storage Service)	AWS EBS (Elastic Block Store)
Storage Type	Object Storage	Block Storage
Attachment	Not directly attached to EC2	Attached to a single EC2 instance
Access	HTTP/HTTPS over the internet	Block-level access by EC2 instance
Use Cases	Unstructured data, backups, archives, static websites	Databases, OS volumes, applications requiring high I/O
Scalability	Virtually unlimited	Scalable within an instance's limits
Performance	High throughput, less emphasis on low latency	Optimized for low-latency, high-performance I/O

Amazon Virtual Private Cloud (VPC)

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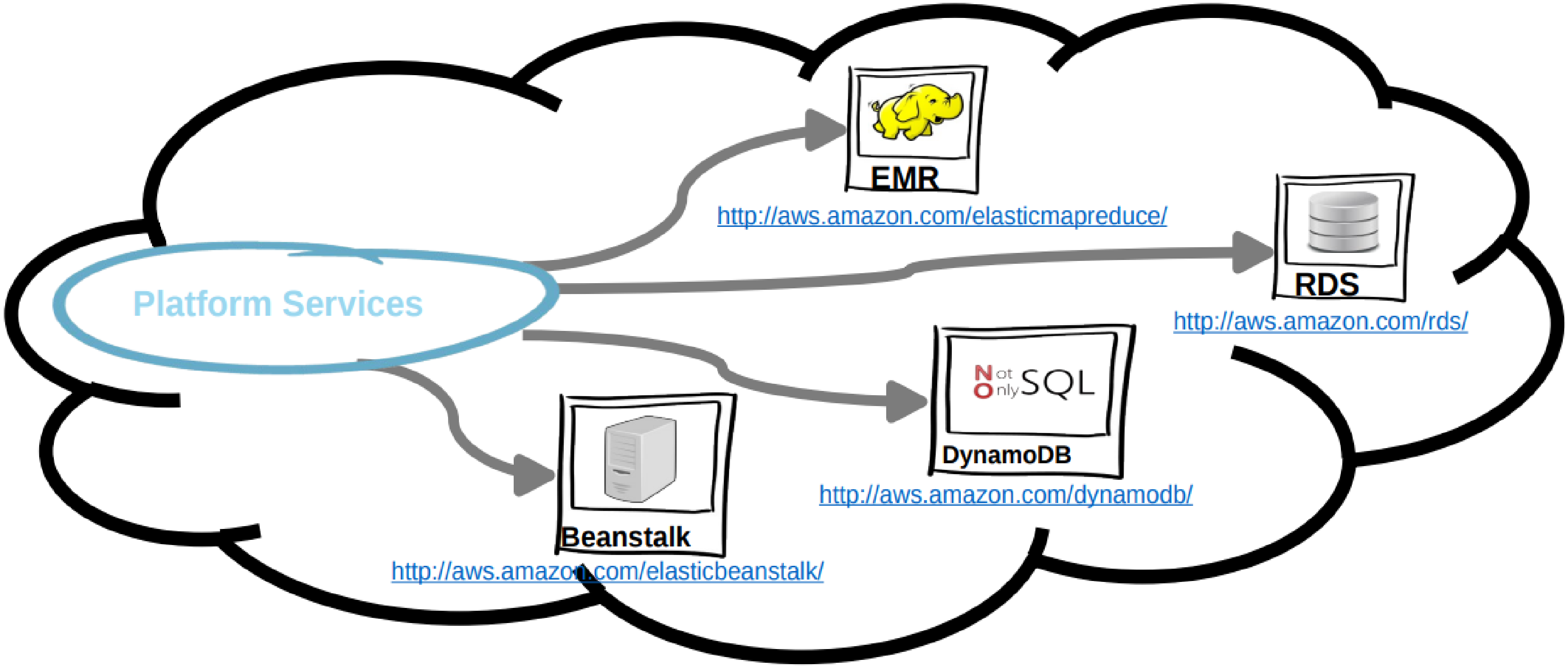
- Amazon VPC lets you provision a logically isolated section of the Amazon Web Services (AWS) Cloud.
- You have complete control over your virtual networking environment, including:

– selection of your own IP address range,
– creation of subnets, and
– configuration of route tables and network gateways.

- VPC allows bridging with an onsite IT infrastructure with an encrypted VPN connection with an extra charge per VPN Connection-hour.
- There is no additional charge for using Amazon Virtual Private Cloud, aside from the normal Amazon EC2 usage charges.

Platform Services



Amazon Elastic MapReduce (EMR)

- Amazon EMR is a web service that makes it easy to quickly and cost-effectively process vast amounts of data using Hadoop.
- Amazon EMR distribute the data and processing across a resizable cluster of Amazon EC2 instances.
- With Amazon EMR you can launch a persistent cluster that stays up indefinitely or a temporary cluster that terminates after the analysis is complete.
- Amazon EMR supports a variety of Amazon EC2 instance types and Amazon EC2 pricing options (OnDemand, Reserved, and Spot).
- When launching an Amazon EMR cluster (also called a "job flow"), you choose how many and what type of Amazon EC2 Instances to provision.
- The Amazon EMR price is in addition to the Amazon EC2 price.
- Amazon EMR is used in a variety of applications, including log analysis, web indexing, data warehousing, machine learning, financial analysis, scientific simulation, and bioinformatics.

Amazon Relational Database Service (RDS)

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- Amazon RDS is a web service that makes it easy to set up, operate, and scale a relational database in the cloud.
- Amazon RDS gives access to the capabilities of a familiar MySQL, Oracle or Microsoft SQL Server database engine.

– Code, applications, and tools already used with existing databases can be used with RDS.

- Amazon RDS automatically patches the database software and backs up the database, storing the backups for a user-defined retention period and enabling point-in-time recovery.
- Amazon RDS provides scaling the compute resources or storage capacity associated with the Database Instance.
- Pay only for the resources actually consumed, based on the DB Instance hours consumed, database storage, backup storage, and data transfer.

– On-Demand DB Instances let you pay for compute capacity by the hour with no long-term commitments.

– Reserved DB Instances give the option to make a low, one-time payment for each DB Instance and in turn receive a significant discount on the hourly usage charge for that DB Instance.

NoSQL Databases

▪ A NoSQL database provides a mechanism for storage and retrieval of data that employs less constrained consistency models than traditional relational databases.

▪ NoSQL databases only support Eventual Consistency which is a consistency model used in distributed computing that informally guarantees that, if no new updates are made to a given data item, eventually all accesses to that item will return the last updated value.

▪ NoSQL databases are often highly optimized key–value stores intended for simple retrieval and appending operations, with the goal being significant performance benefits in terms of latency and throughput.

▪ Key–value stores allow the application to store its data in a schema-less way.

– The data could be stored in a datatype of a programming language or an object. Because of this, there is no need for a fixed data model

Amazon DynamoDB

▪ DynamoDB is a fast, fully managed NoSQL database service that makes it simple and cost-effective to store and retrieve any amount of data, and serve any level of
request traffic.

▪ All data items are stored on Solid State Drives (SSDs), and are replicated across 3Availability Zones for high availability and durability.

▪ DynamoDB tables do not have fixed schemas, and each item may have a different number of attributes.

▪ DynamoDB has no upfront costs and implements a pay as you go plan as a flat hourly rate based on the capacity reserved.

Amazon Elastic Beanstalk

- AWS Elastic Beanstalk provides a solution to quickly deploy and manage applications in the AWS cloud.
- You simply upload your application, and Elastic Beanstalk automatically handles the deployment details of capacity provisioning, load balancing, auto-scaling, and application health monitoring.
- Elastic Beanstalk leverages AWS services such as Amazon EC2, Amazon S3, ….
- To ensure easy portability of your application, Elastic Beanstalk is built using familiar software stacks such as:

– Apache HTTP Server for Node.js, PHP and Python
– Passenger for Ruby,
– IIS 7.5 for .NET
– Apache Tomcat for Java.

- There is no additional charge for Elastic Beanstalk - you pay only for the AWS resources needed to store and run your applications.

Amazon Simple Queue Service (SQS)

Amazon Simple Queue Service (SQS) is **a message queuing service**.

It enables you to **decouple and scale** microservices, distributed systems, and server less applications.

SQS **eliminates** the complexity and overhead associated with managing and operating message oriented middleware.

Using SQS, you can **send, store, and receive messages** between software components at any volume.

SQS offers two types of message queues.

Standard queues offer maximum throughput, best-effort ordering, and at-least-once delivery.

SQS FIFO queues are designed to guarantee that messages are processed exactly once, in the exact order that they are sent.

Unlimited queues and messages.

Payload Size: Message payloads can contain up to **256KB** of text in any format.

Batches: Send, receive, or delete messages **in batches** of up to 10 messages or 256KB.

Batches cost the same amount as single messages, meaning SQS can be even **more cost effective** for customers that use batching.

Long polling: Reduce **extraneous polling** to minimize cost while receiving new messages as quickly as possible.

Retain messages in queues for up to **14 days**.

Send and receive messages **simultaneously**.

Message locking: When a message is received, it becomes “locked” while being processed. This keeps other computers from processing the message simultaneously.

Queue sharing: Securely share Amazon SQS queues anonymously or with specific AWS accounts.

Server-side encryption (SSE): Protect the contents of messages in Amazon SQS queues using keys managed in the AWS Key Management Service (AWS KMS). SSE encrypts messages as soon as Amazon SQS receives them. The messages are stored in encrypted form and Amazon SQS decrypts messages only when they are sent to an authorized consumer.

Dead Letter Queues (DLQ): Handle messages that have not been successfully processed by a consumer with Dead Letter Queues.

Launching an Amazon SQS service for the first time need the following steps:

Sign up for AWS

Create an IAM user

Get your access key ID and secret access key

Create a Queue

Send a Message

Receive a Message

Delete a Message

Delete a Queue

To create an AWS account

Open https://portal.aws.amazon.com/billing/signup.

Follow the online instructions.

Create an IAM user with administrative privilege.

Sign in to the Amazon SQS console.

Choose **Create New Queue.**

On the **Create New Queue** page, ensure that you're in the correct region and then type the **Queue Name**. The name of a FIFO queue must end with the ***.fifo*** suffix.

Standard is selected by default.

To create your queue with the default parameters, choose **Quick-Create Queue**.

Your new queue is created and selected in the queue list.

The **Queue Type** column helps you distinguish standard queues from FIFO queues at a glance.

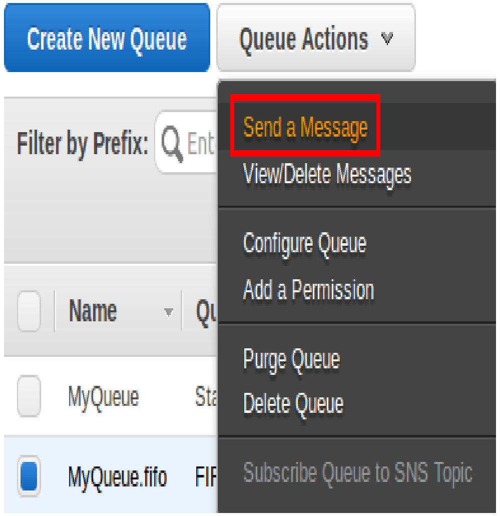
Your queue's **Name**, **URL**, and **ARN** are displayed on the **Details** tab.

Send a Message

From the queue list, select the queue that you've created.

From Queue Actions, select Send a Message.

The **Send a Message** to QueueName dialog box is displayed.



View/Delete Messages

From the queue list, **select** the queue that you have created.

From **Queue Actions**, select View/Delete Messages.

Choose **Start Polling** for messages.

Amazon SQS begins to poll the messages in the queue. The dialog box displays a message from the queue. A progress bar at the bottom of the dialog box displays the status of the message's visibility timeout.

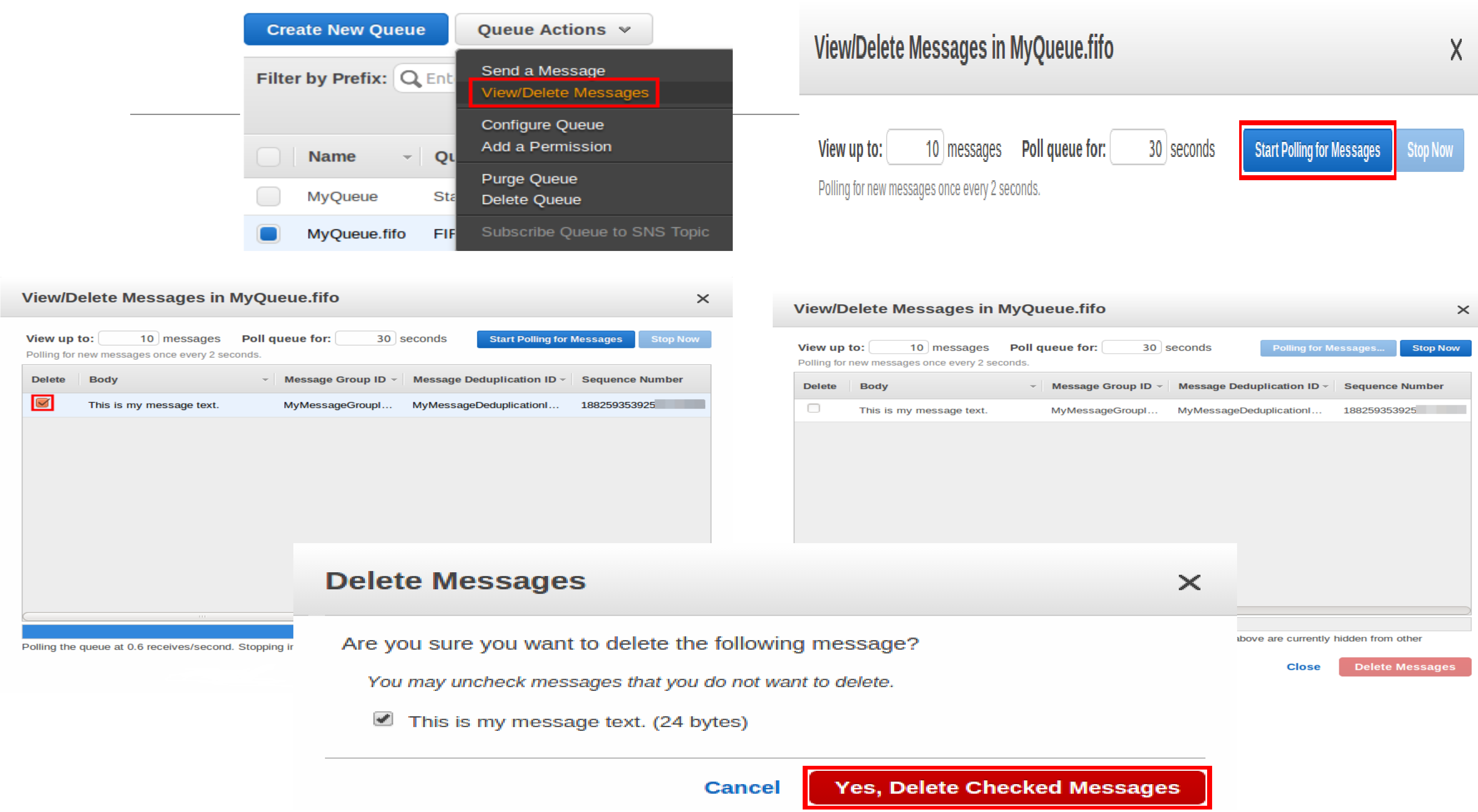
When the progress bar is filled the message becomes visible to consumers.

Before the visibility timeout expires, select the message that you want to delete and then choose Delete Message.

In the **Delete Messages** dialog box, confirm that the message you want to delete is checked and choose Yes, Delete Checked Messages.

The selected message is deleted.

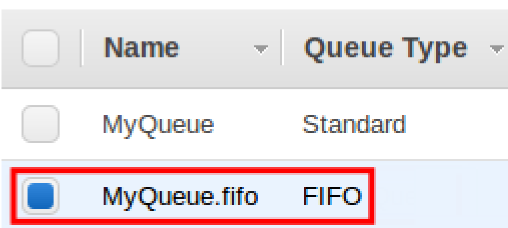
Select Close.

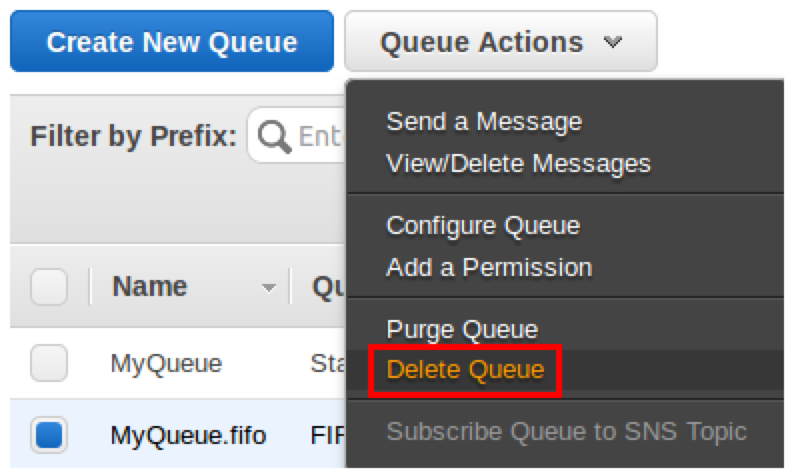

Create New Queue
Queue Actions
Filter by Prefix:
Send a Message
View/Delete Messages
Configure Queue
Add a Permission
Purge Queue
Delete Queue
Subscribe Queue to SNS Topic
Name
MyQueue
MyQueue.fifo
View/Delete Messages in MyQueue.fifo
View up to: 10 messages Poll queue for: 30 seconds
Start Polling for Messages
Stop Now
Polling for new messages once every 2 seconds.
Delete
Body
Message Group ID
Message Deduplication ID
Sequence Number
This is my message text.
MyMessageGroupI...
MyMessageDeduplicationI...
188259353925
Polling for Messages...
Polling the queue at 0.6 receives/second. Stopping i
Close
Delete Messages
Delete Messages
Are you sure you want to delete the following message?
You may uncheck messages that you do not want to delete.
This is my message text. (24 bytes)
Cancel
Yes, Delete Checked Messages

Delete Queue.

From the queue list, select the queue that you have created.

From Queue Actions, select Delete Queue.

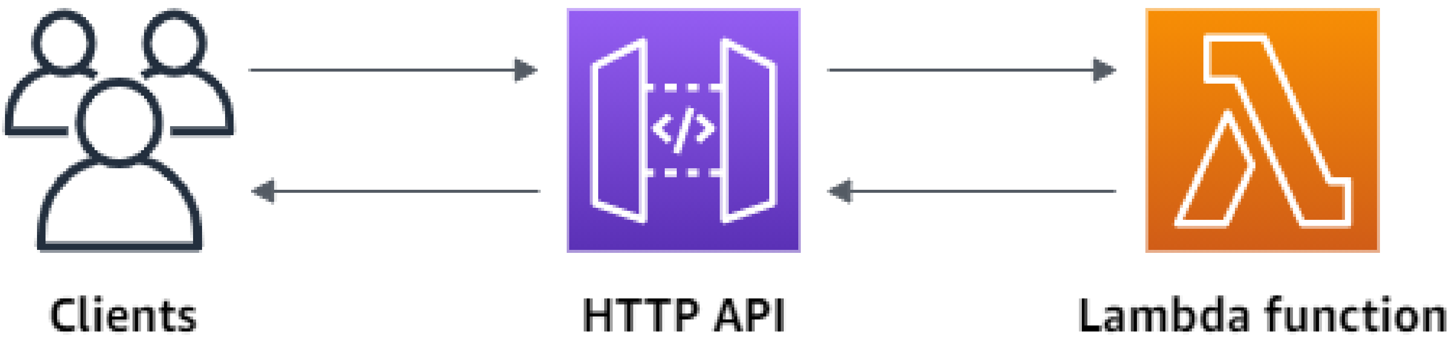




Getting started with API Gateway

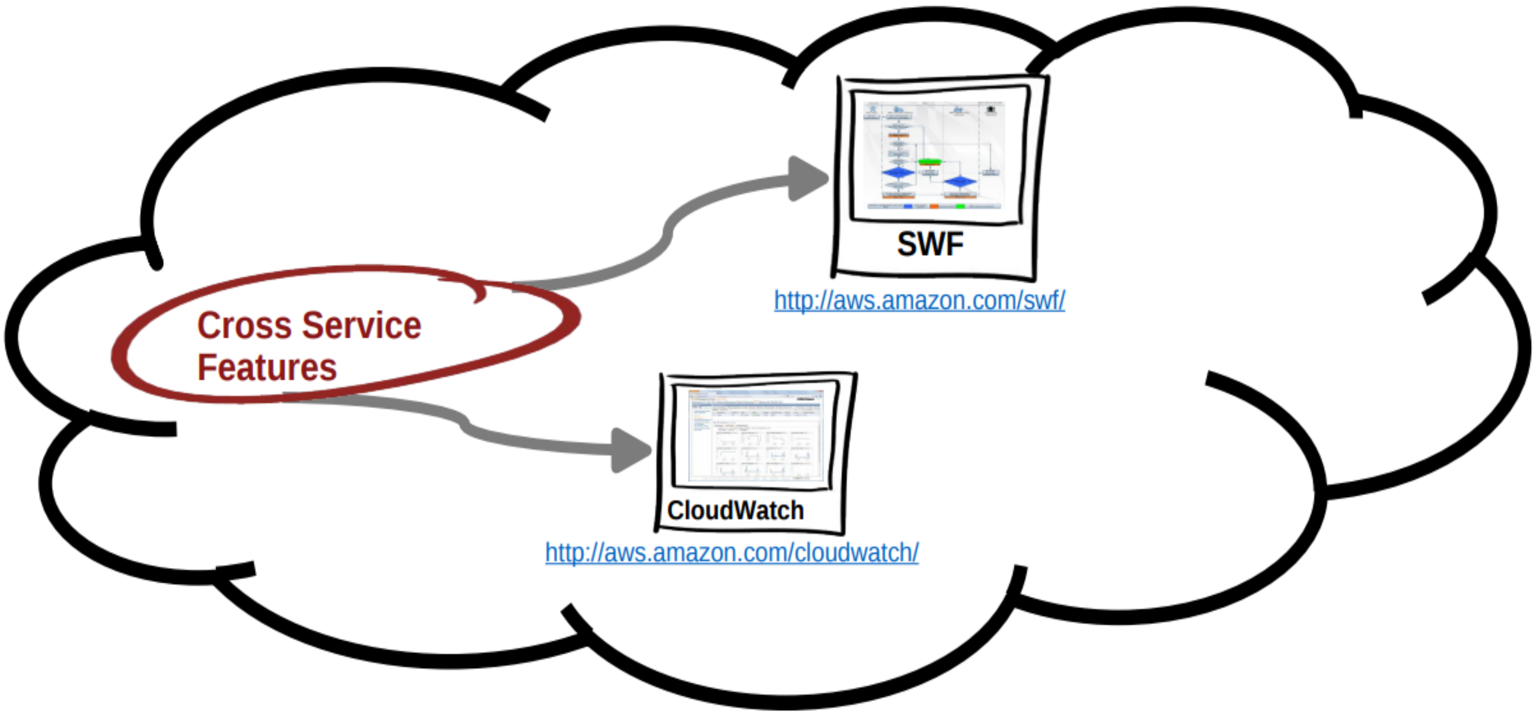
You create a serverless API.
Serverless APIs let you focus on your applications, instead of spending time provisioning and managing servers.

Step 1: Create a Lambda function
Step 2: Create an HTTP API
Step 3: Test your API
(Optional) Step 4: Clean up



When you invoke HTTP API,
API Gateway routes the request to Lambda function.
Lambda runs the Lambda function and returns a response to API Gateway.
API Gateway then returns a response to you.

Cross Service Features



Amazon CloudWatch

- Amazon CloudWatch provides monitoring for AWS cloud resources and the applications customers run on AWS.
- Amazon CloudWatch lets you programmatically retrieve your monitoring data, view graphs, and set alarms to help you troubleshoot, spot trends, and take automated action based on the state of your cloud environment.
- Amazon CloudWatch enables you to monitor your AWS resources up-to-the-minute in real-time, including:

– Amazon EC2 instances,
– Amazon EBS volumes,
– Elastic Load Balancers,
– Amazon RDS DB instances.

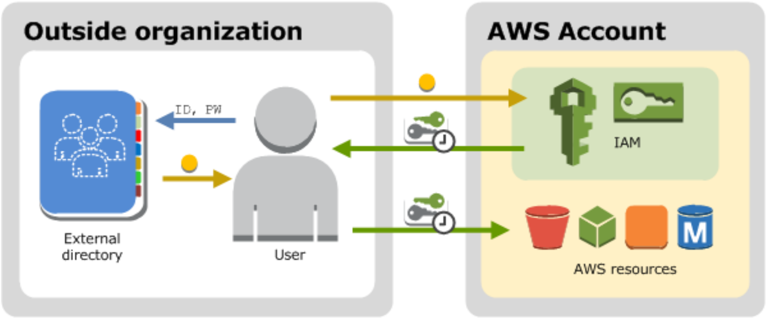
- Metrics such as CPU utilization, latency, and request counts are provided automatically for these AWS resources.
- Customers can also supply their own custom application and system metrics, such as memory usage, transaction volumes, or error rates.

Amazon Simple Workflow Service (SWF)

- Amazon SWF is a task coordination and state management service for cloud applications.
- Using Amazon SWF, you structure the various processing steps in an application that runs across one or more machines as a set of "tasks."
- Amazon SWF manages dependencies between the tasks, schedules the tasks for execution, and runs any logic that needs to be executed in parallel.
- The service also tracks the tasks' progress.
- As the business requirements change, Amazon SWF makes it easy to change application logic without having to worry about the underlying state machinery and flow control.

AWS Identity and Access Management (IAM)

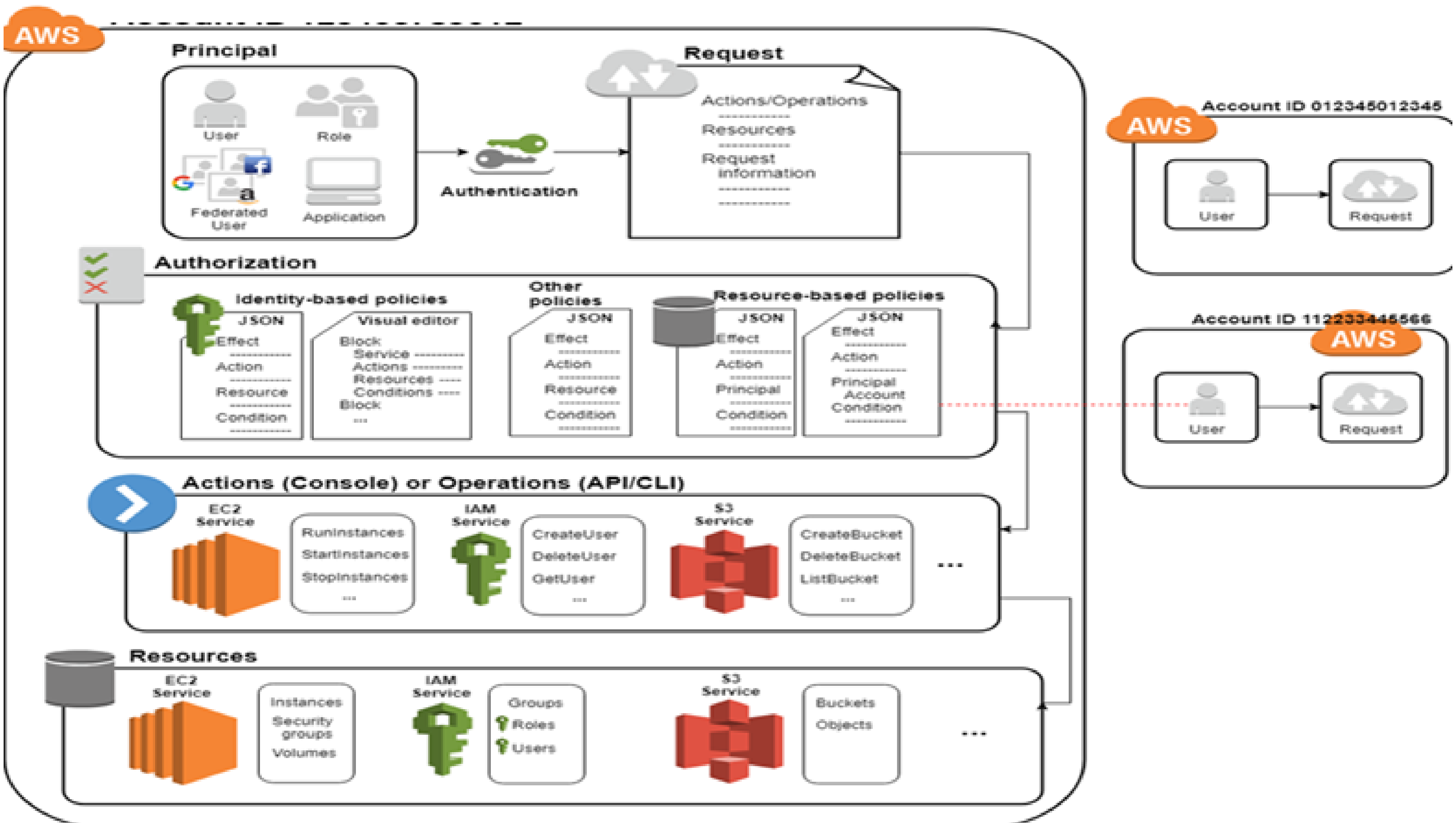
- AWS Identity and Access Management (**IAM**) enables you to manage access to AWS services and resources securely.
- Using IAM, one can create and manage AWS users and groups, and use permissions to allow and deny their access to AWS resources.
- IAM is a feature of AWS account offered at no additional charge.
- To get started using IAM use the AWS Management Console.



IAM Features

- Allow user to administer and use resources in you AWS account without sharing password.
- Grant permission to different people for different resources.
- Can provide application usage restriction on EC2.
- You can use access key to make any request to AWS.
- IAM gives high availability and consistency across the globe.
- Allow two-factor authentication to the user for additional security.
- Allow AWS CloudTrail to view IAM identities who made request to resources in your account.

IAM Elements



IAM-Principal

- A *principal* is a person or application that can make a request for an action or operation on an AWS resource.
- The principal is authenticated as the AWS account root user or an IAM entity to make requests to AWS.

IAM-Request

- When a principal tries to use the AWS Management Console, it sends a *request* to AWS. The request includes the following information:
  - **Actions or operations** – The actions or operations that the principal wants to perform.
  - **Resources** – The AWS resource object upon which the actions or operations are performed.
  - **Principal** – The person or application that used an entity (user or role) to send the request.
  - **Environment data** – Information about the IP address, user agent, SSL enabled status, or the time of day.
  - **Resource data** – Data related to the resource that is being requested.
- AWS gathers the request information into a *request context*, which is used to evaluate and authorize the request.

IAM-Authentication

- A principal must be authenticated (signed in to AWS) using their credentials to send a request to AWS.
- To authenticate from the console as a root user, you must sign in with your email address and password.
- As an IAM user, provide your account ID or alias, and then your user name and password. To authenticate from the API or AWS CLI, you must provide your access key and secret key.
- During authorization, AWS uses values from the request context to check for policies that apply to the request.
- It then uses the policies to determine whether to allow or deny the request.
- AWS checks each policy that applies to the context of your request. If a single permissions policy includes a denied action, AWS denies the entire request and stops evaluating. This is called an *explicit deny*.
- An *explicit allow* in any permissions policy (identity-based or resource-based) overrides this default policies.

IAM-Actions or Operations

- After your request has been authenticated and authorized, AWS approves the actions or operations in your request.
- Operations are defined by a service, and include things that you can do to a resource, such as viewing, creating, editing, and deleting that resource.
- For example, IAM supports approximately 40 actions for a user resource, including actions like CreateUser, DeleteUser, GetUser, UpdateUser.
- To allow a principal to perform an operation, you must include the necessary actions in a policy that applies to the principal or the affected resource.

IAM-Resources

A resource is an object that exists within a service.
Examples include an Amazon EC2 instance, an IAM user, and an Amazon S3 bucket.
The service defines a set of actions that can be performed on each resource

AWS Free Usage Tier

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http://aws.amazon.com/free/

https://docs.aws.amazon.com/index.html